

Sample Paper 7 — Answer Key (Level-2 Difficulty Step-Up)

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **C) Carrot** — Mango, Banana, and Apple are fruits; Carrot is a vegetable.
2. **A) Music** — An author creates a book; a composer creates music.
3. **A) PDWKV** — The rule is "+3 letters" (wrapping Z $\rightarrow$ C): M $\rightarrow$ P, A $\rightarrow$ D, T $\rightarrow$ W, H $\rightarrow$ K, S $\rightarrow$ V.
4. **C) 65** — Two patterns are interleaved: the odd-position terms go 3, 7, 11, 15 (+4 each); the even-position terms go 80, 75, 70, ? (–5 each) — so the next even term is 65.
5. **A) Omkar, Leena, Madhav, Nisha, Kabir** — Madhav is at position 3 (middle). Omkar is at the left end, position 1. Nisha is immediately right of Madhav, position 4. Leena sits between Omkar (1) and Madhav (3), so Leena is position 2. The only person left, Kabir, takes position 5.
6. **A) Back at the starting point** — East and West cancel (11 m each way); North and South cancel (8 m each way).
7. **B) Tuesday** — $29 - 4 = 25$ days later; $25 = 3$ weeks + 4 days, so the day of the week moves forward 4 days from Friday: Tuesday.
8. **D) 90** — $8 = 2 \times 2 \times 2$, $27 = 3 \times 3 \times 3$, $64 = 4 \times 4 \times 4$ (each a number multiplied by itself twice — a "cube number"); 90 does not fit this pattern.
9. **C) 360** — Each term is multiplied by one more than the last multiplier used ($\times 2$, $\times 3$, $\times 4$, $\times 5$): $72 \times 5 = 360$.
10. **B) 6th** — Being 9th from the left means 8 students are to Neha's left. The 3 students who leave from the left end are among those to her left, so her position from the left decreases by 3: $9 - 3 = 6$ th.
11. **C) 42** — 17, 31, and 53 are prime numbers; 42 ($= 2 \times 3 \times 7$) is not.
12. **A) ICTFGP** — The rule is "+2 letters": G $\rightarrow$ I, A $\rightarrow$ C, R $\rightarrow$ T, D $\rightarrow$ F, E $\rightarrow$ G, N $\rightarrow$ P.
13. **D) Sphere** — Square, Rhombus, and Trapezium are flat (2-D) shapes; a Sphere is a solid (3-D) shape.
14. **D) Vikas** — The finishing order is Vikas, then Tarun, then Rohan, then Suresh, so Vikas finished first.
15. **B) 17** — The gaps between terms are 1, 2, 3, 4 — each one more than the last. The next gap is 5: $12 + 5 = 17$.

Section 2 — Mathematical Reasoning

16. **A) 9,13,568** — $5,67,890 + 3,45,678 = 9,13,568$.
17. **A) 3,82,110** — $9,50,000 - 5,67,890 = 3,82,110$.
18. **A) 91,848** — $7,654 \times 12 = 91,848$.
19. **A) 4,567** — $1,09,608 \div 24 = 4,567$ exactly.
20. **C) 42** — $84 = 2^2 \times 3 \times 7$, $126 = 2 \times 3^2 \times 7$. HCF = $2 \times 3 \times 7 = 42$.
21. **C) 72** — $18 = 2 \times 3^2$, $24 = 2^3 \times 3$. LCM = $2^3 \times 3^2 = 72$.
22. **D) 13/15** — As decimals: $6/7 \approx 0.857$, $9/11 \approx 0.818$, $5/6 \approx 0.833$, $13/15 \approx 0.867$. The greatest is 13/15.
23. **C) 6 km 800 m** — $3850 \text{ m} + 2950 \text{ m} = 6800 \text{ m} = 6 \text{ km } 800 \text{ m}$.
24. **B) 1,092 sq m** — Area = $42 \times 26 = 1,092 \text{ sq m}$.
25. **C) 12 cm** — A regular decagon has 10 equal sides: $120 \div 10 = 12 \text{ cm}$.
26. **C) 30%** — Profit = $1,950 - 1,500 = ₹450$; profit % = $450/1500 \times 100 = 30\%$.
27. **C) 20%** — If principal P doubles, the interest earned equals P itself. $SI = (P \times R \times T) \div 100 \rightarrow P = (P \times R \times 5) \div 100 \rightarrow R = 100 \div 5 = 20\%$.
28. **D) ₹210** — Total parts = $4+5+7 = 16$; 1 part = $480 \div 16 = 30$; largest share = $7 \times 30 = ₹210$.
29. **A) 3/8** — $0.375 = 375/1000$, which simplifies to 3/8.
30. **C) 12 kg** — $5250 \text{ g} + 3800 \text{ g} + 2950 \text{ g} = 12000 \text{ g} = 12 \text{ kg}$.
31. **C) 12,600** — $3.5 \text{ hours} \times 3600 \text{ seconds/hour} = 12,600 \text{ seconds}$.
32. **B) 93 cm** — An equilateral triangle has 3 equal sides: $3 \times 31 = 93 \text{ cm}$.
33. **B) 1,134 sq cm** — Area = $42 \times 27 = 1,134 \text{ sq cm}$.
34. **D) 12** — A cuboid has 12 edges (4 on top, 4 on the bottom, 4 vertical).
35. **C) 14** — Number of diagonals = $n(n-3)/2 = 7 \times 4/2 = 14$.

Section 3 — Everyday Mathematics

36. **C) ₹25,000** — $2,25,000 \div 9 = 25,000$.
37. **C) 2,100** — Rate = $2,800 \div 8 = 350 \text{ bottles/hour}$; in 6 hours = $350 \times 6 = 2,100$.
38. **C) 25 min** — $4,500 \div 180 = 25 \text{ minutes}$.
39. **C) 4 h 23 min** — Total time 6:55 a.m. to 11:40 a.m. = 4 h 45 min; subtracting the 22-minute stop gives 4 h 23 min of actual travel.
40. **C) 1 L 400 mL** — $4800 \text{ mL} + 2200 \text{ mL} = 7000 \text{ mL}$; $7000 \div 5 = 1400 \text{ mL per bottle}$.
41. **C) ₹284** — $9 \times ₹35 = ₹315$; $₹315 - ₹31 \text{ discount} = ₹284$.
42. **C) 1,120 L** — $1/10 \times 1,600 = 160 \text{ L used per day}$; over 3 days = 480 L used; $1,600 - 480 = 1,120 \text{ L remaining}$.
43. **C) 63 kg** — $19500 \text{ g} + 17750 \text{ g} + 14250 \text{ g} + 11500 \text{ g} = 63,000 \text{ g} = 63 \text{ kg}$.

44. **C) 48** — $1,680 \div 35 = 48$ rows.

45. **C) ₹540** — Sold for $90 \times ₹46 = ₹4,140$; profit = $₹4,140 - ₹3,600 = ₹540$.

Section 4 — Achievers Section (3 marks each)

46. **C) 22** — Working backwards: $(126 + 28) \div 7 = 154 \div 7 = 22$.

47. **C) 36** — Let the first number be x . Then second = $2x$, third = $2x+20$. Sum: $x + 2x + (2x+20) = 200$
 $\rightarrow 5x + 20 = 200 \rightarrow x = 36$.

48. **C) ₹1,920** — After spending $3/8$ on books, $5/8$ of the money remains. After spending $1/6$ of that on stationery, $5/6$ of $5/8 = 25/48$ of the original remains. So $25/48$ of the total = ₹1,000 $\rightarrow$ total = ₹1,920.

49. **D) 214** — Girls = $(2 \times 104) + 6 = 214$. Check: $104 + 214 = 318$. ✓

50. **C) 55** — Each term is the sum of the previous two: $21 + 34 = 55$.

Scoring guide (out of 60, Level-2-difficulty paper): 32+ here is a very strong result — it's harder than Papers 1–6 by design, so the usual 42+ Level-1 benchmark doesn't directly apply. 22–31 is solid for a paper at this difficulty. Below 22 — review *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.
